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Neural network metamodelling in multi-objective optimization of a high latitude solar community

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A B S T R A C T

A solar community of 100 residential houses was optimized for Finnish conditions with the aim of achieving a 90% solar fraction for both space heating and domestic hot water. Optimization was done using a novel method based on neural network metamodelling and compared to the standard NSGA-II genetic algorithm. Compared to NSGA-II, the new method obtained a larger hypervolume by finding better solutions both in the center and edge of the non-dominated front. The combined non-dominated front of both methods was better than either one separately. The performance target was achieved as the optimal solar community designs had heating solar fractions ranging from 64% to 95%.

1. Introduction

Global solar energy capacity continues to increase (van der Hoeven, 2014). Unfortunately, a large penetration of solar generation capacity may cause problems with the electric grid due to the diurnal and undispatchable nature of solar energy (Denholm and Hand, 2011). Seasonal variability of solar energy is another major problem, which is especially significant in high latitudes, such as in the Nordic countries. To combat this problem, seasonal energy storage is needed. Except for pumped hydro facilities, large scale seasonal electricity storage is currently not economically feasible (Beaudin et al., 2010; Pierpoint, 2016). Thermal storage, however, is already a mature technology ready for deployment (Navarro et al., 2016).

A typical thermal storage system is a hot water tank in a house, for which the optimal size has been investigated before (Rodríguez-Hidalgo et al., 2012). Such a system can even be used to store excess electricity from solar panels to a limited degree (Hirvonen et al., 2016). Tanks are usually only used for short-term storage and additional systems may be needed for seasonal storage. Underground tank-based seasonal storage was used in the first practical Finnish solar community study (Lund, 1984), though the storage turned out to be undersized. Different storage technologies have been reviewed in Xu et al. (2014) and many real-life projects summarized in Schmidt et al. (2004) and Bauer et al. (2010)). One increasingly common seasonal storage type is the borehole thermal energy storage (BTES), where the heat is stored directly to the ground through boreholes drilled into rock or soil (Rad and Fung, 2016). Thermal storage efficiency scales with size, as heat losses get relatively smaller with increasing storage volume. Because energy generation capacity also gets cheaper with size, seasonal storage systems should be community solutions. A successful example is the Drake Landing Solar Community in Canada (Sibbitt et al., 2011), which has achieved a 98% solar fraction for space heating (SH) through seasonally stored solar energy, providing heat for 52 residential houses.

Such systems cannot be directly copied to other locations, but need to be adjusted to the local conditions (Flynn and Sirén, 2015). Properties such as ground conductivity and thermal capacity affect system performance. In this study, the Drake Landing design is modified to supply both space heating and domestic hot water (DHW) for a community in Finland (above 60°latitude). Due to the high latitude, the winter heating demand is high while seasonal solar variability is very large. There is practically no solar energy available during winter, making seasonal storage mandatory to achieve high solar fraction in heating. To help achieve higher temperatures required for DHW, a solar assisted ground source heat pump is included in the energy system. To maximize

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Nomenclature

Abbreviation	Explanation
ANN	artificial neural network
BTES	borehole thermal energy storage
DHW	domestic hot water
G	total incident solar radiation
HP	heat pump
LCC	life cycle cost
NSGA	non-dominated sorting genetic algorithm
PV	photovoltaics
SF	solar fraction
SH	space heating
SPF	seasonal performance factor
ST	solar thermal

the feasibility of the design, multi-objective optimization is used to minimize both external energy demand and life cycle cost (LCC).

Many different methods exist for energy system optimization. Recent storage-integrating district heating optimization studies have been reviewed in Olsthoorn et al. (2016). Typical single-objective optimization problems are about minimization of cost or energy/exergy consumption, often solved using mixed integer linear programming (MILP). For multi-objective optimization, genetic algorithms such as NSGA-II (Deb et al., 2002) are a common choice. A genetic algorithm was used to optimize the energy system of a pre-simulated zero energy building in Ascione et al. (2016) and also for optimizing the seasonal storage and solar heating system of a greenhouse (Durão et al., 2014). Other population-based methods such as particle swarm optimization have also been used for sizing solar power installations (Khare and Rangnekar, 2013). Simulated annealing has also been used for the same purpose (Ekren and Ekren, 2010). Linear programming with sparse revised Simplex was used to optimize the design for a hybrid renewable energy community (Wang et al., 2015). Interval-linear programming and chance-constrained programming were combined in Cai et al. (2009) to perform optimization under uncertainties, such as solar and wind availability. Similar interval-based optimization was used for scheduling optimization of renewable energy systems (Chen et al., 2015).

In simulation-based optimization, the actual simulations can often take a long time compared to other processes in the optimization. This makes it very important to minimize the number of simulations required. One method to achieve this goal is to do optimization in separate stages (Hamdy et al., 2013). This way, unreasonable variable combinations can be filtered out and partially optimized results used as inputs. Also, if some part of a system model is independent of the others, it may be possible to simulate it only one time and just use tabulated values for the rest of the runs, reducing the need for time-consuming simulations.

Another way to reduce the need for simulations is through metamodelling (or surrogate models) (Bornatico et al., 2013). A metamodel is a model of a model, meaning that it is generated out of output from another model. Metamodels are typically very fast to run compared to detailed simulation programs. Utilizing a metamodel can reduce function evaluation time from minutes or hours to a fraction of a second. One metamodelling technique is the use of artificial neural networks (ANN). The principles of ANN are explained in detail in Kalogirou (2000). In short, neural networks use linear equations to find relations between inputs and outputs and can be used to quickly simulate even phenomena that are not completely understood, as long as proper training data is available.

For example, ANN was used to make a metamodel out of a TRNSYS building model, which was then used with a genetic algorithm (GA) to do optimization (Asadi et al., 2014). First, parametric runs were done to generate a database for training the ANN. The ANN metamodel is very fast to use and allows quick optimization. Unfortunately, generating an adequate training set to ensure the accuracy of the metamodel requires a significant amount of simulations, so actual time saving compared to direct optimization is not guaranteed. Generally neural networks are trained with an extensive dataset and then used to completely replace the original model to complete optimization, as was done in Boukelia et al. (2016) and Magnier and Haghighat (2010). However, other methods such as kriging have also been used for parallel optimization and surrogate modeling (Hussein and Deb, 2016).

The novelty in this study is the introduction of a new neural network based optimization method for problems where the function evaluation takes a long time compared to other calculation processes. Instead of generating a complete training set before optimization, the optimization is started with a very small training set which is updated as the optimization progresses. The ANN is retrained as new samples are added after each optimization step, increasing the ANN accuracy and improving optimization results. Additionally, instead of a single neural network for the whole search space, several neural networks are trained in parallel, each representing a different section of the objective space. The optimization method will be applied to a solar community design problem, which will reveal new information about solar heating and seasonal energy storage in high latitude Nordic conditions in the community scale. The objective is to find a range of energy-economic optima for solar communities with a high solar fraction. An important subobjective is to find a configuration with over 90% solar fraction for heating.

2. Materials and methods

2.1. Energy system details and modeling

The study centers on a hypothetical Finnish solar community with a local heating grid, modeled using TRNSYS 17 and first introduced in ur Rehman et al. (2016). The solar community consisted of a 100 residential buildings, each with a heated area of 100 m^2. The design of the centralized heating system is shown in Fig. 1. Heat generated by the solar thermal collectors was stored in either the high temperature domestic hot water (DHW) tank at around 60 °C or the low temperature space heating (SH) tank at around 40 °C, depending on tank temperature levels and the chosen control algorithm. If the tank temperature rose 10 °C above the setpoint, the energy in the tank was discharged into the seasonal storage until the tank had cooled down enough.

The seasonal storage was a borehole thermal energy storage (BTES), which is a grid of boreholes drilled into the ground. Each borehole was fitted with U-tube piping which served as a heat exchanger between the ground and heat transfer fluid. Several boreholes could be connected in series so that the fluid exiting from one borehole could be pumped into the next one. In charging mode, hot fluid was pumped into the center of the storage and the cooled output flow was directed into the next borehole in series. Thus, a radial temperature distribution could be formed, where the center of the storage had the highest temperature and the edges had the lowest, minimizing heat losses to the surroundings.

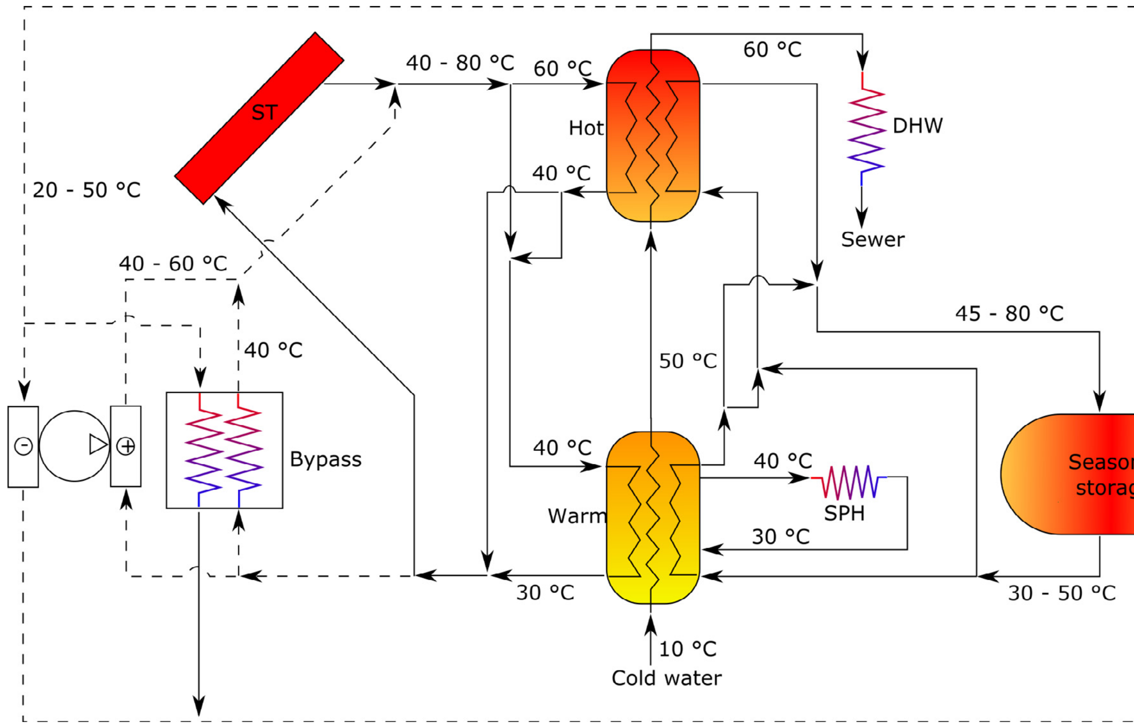


Fig. 1. The heating system consists of solar thermal collectors, two buffer tanks, a borehole seasonal energy storage and a heat pump. Parallel connection allows solar collectors to charge either tank separately, while series connection treats them as a single tank.

When the tank temperature dropped too low, the BTES was discharged back into the tanks. In this mode, the cold fluid entered from the cool outer edge and exited from the hot center. If the BTES output temperature was high enough, it could be directly utilized for heating the tanks. If the BTES output temperature was low or the energy demand is high, the BTES flow was used as the energy source for heat pumps (HP). Because the ground temperature was often higher than typically used with a ground-source heat pump (GSHP), the coefficient of performance (COP) of the HP is improved (Fig. 2), going as high as 8 for low temperature heat (for SH need). Heat generated by the evaporator component of the heat pump was considered to be solar energy, while heat generated by the HP compressor was not.

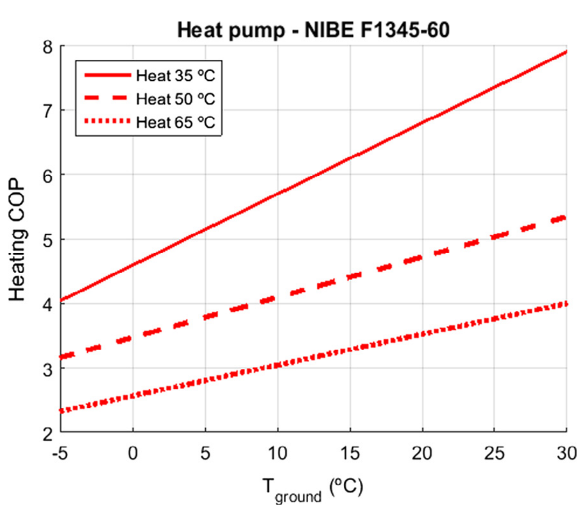


Fig. 2. The COP of the heat pump increases until the source temperature reaches 30 °C. Above that point the inlet flow will be mixed with cooled output fluid and COP remains constant.

If the heat pump and solar energy were not enough to meet the temperature needs, backup heating was handled by direct electric heaters. The tanks could be connected to the solar collectors in series or in parallel and the output temperature of the collectors was determined by the connection and controlled through flow rate adjustments. If both tanks were at adequate temperature levels, all solar heat was pumped into the lower temperature tank to maximize energy efficiency. Cold DHW flow was preheated in the low temperature tank before the final heating to temperatures above 55 °C in the hot tank. This maximized the use of low grade heat, improving total system efficiency. Cooling needs in the community were so small that a cooling system was not included.

Weather data used for simulation was the Helsinki Test Reference Year 2012 (Kalamees et al., 2012), with annual solar insolation of 975 kW h/m^2. In Finland the average thermal conductivity of the ground (3.50 W/mK) is higher than in the Drake Landing Solar Community, which increases losses, while the volumetric heat capacity is lower (2240 kJ/m^3 K), reducing storage capacity (Flynn and Sirén, 2015). This makes high performance more difficult to achieve.

The solar fraction was defined separately for heating and total electricity demand. For heating, it was defined using the ratio of electricity needed to run the heating system vs. the total heat demand met by the system

$$SF_h = 1 - \frac{E_{heating}}{E_{heat,dem}} = 1 - \frac{E_{pumps} + E_{HP} + E_{backup}}{E_{SH} + E_{DHW}}, \tag{1}$$

where $E_{heating}$ is the electricity needed to run the heating system, $E_{heat,dem}$ is the total heating energy used in the buildings, E_{pumps}, E_{HP} and E_{backup} are the electricity use of the pumps, heat pump and backup heating, respectively and E_{SH} and E_{DHW} are the actual amount of heat used to provide space heating and domestic hot water.

Since grid electricity was the only external energy source, the solar fraction for the whole system was defined using the ratio of imported electricity vs. the total electricity demand

$$SF_t = 1 - \frac{E_{\text{elec,import}}}{E_{\text{elec,dem}}} = 1 - \frac{E_{\text{heating}} + E_{\text{appliances}} - E_{\text{self-consumption}}}{E_{\text{heating}} + E_{\text{appliances}}}, \quad (2)$$

where $E_{\text{elec,import}}$ is the electricity imported from the grid, $E_{\text{elec,dem}}$ is the total electricity use by the buildings, $E_{\text{appliances}}$ is the energy demand of lights and other non-heating appliances and $E_{\text{self-consumption}}$ is the amount of locally generated solar electricity that is used on-site (not exported as excess).

The seasonal performance factor for heating was the average amount of useful heat generated by each unit of electricity and was defined as

$$SPF = \frac{E_{\text{heat,dem}}}{E_{\text{heating}}} = \frac{E_{\text{SH}} + E_{\text{DHW}}}{E_{\text{pumps}} + E_{\text{HP}} + E_{\text{backup}}}. \quad (3)$$

The life cycle cost included operation costs (cost of imported energy) from 25 years. Because the BTES system required several years to heat up and achieve optimal performance, it was not enough to run the simulation for just one year, but due to long simulation time, 25 year simulation was also not feasible. As a compromise, the system was simulated for 5 years and the final year was used for estimating the cost and performance for all the remaining years.

The prices used for system components are listed in Table 1. Generally the high prices were for small amounts or capacities and the low prices for large amounts or capacities. For some components, a single constant price was used. Component prices include the labor cost, which was 20–30% of the total. When determining building space heating demand, separate simulations were run to find Pareto optimal combinations of windows, insulation and heat recovery systems. These choices were then used as the building quality component. The base case building was designed to meet Finnish building requirements and had a cost of 0, because the basic building was seen as an unavoidable cost separate from the energy system. The cost of the higher quality buildings was given as a difference to the base case and added to the energy system cost. The base case building had a space heating demand of 50 kW h/m^2/a, while the highest quality building had a demand of 25 kW h/m^2/a. All buildings had a DHW demand of 35 kW h/m^2/a and an appliance electricity demand of 40 kW h/m^2/a.

2.2. Optimization problem

The optimization problem was defined as

$$\begin{aligned} &\min\{f_1(x), f_2(x)\} \\ &\text{s.t. } g(x) = A_{ST} + A_{PV} \leqslant 60 \text{ m}^2 \\ &lb_i \leqslant x_i \leqslant ub_i, i \in \{1, \ldots, 14\}, \end{aligned} \quad (4)$$

where f_1 is the amount of annually imported energy as primary energy equivalent, f_2 is the life cycle cost, g is the limit on roof space and lb_i and ub_i are the lower and upper bounds for all decision variables. The decision variables are introduced in Table 2. Primary energy was calculated using the primary energy factor of 1.7 for grid electricity, which was the only external energy source (Finlex, 2013). The life cycle cost was the sum of initial investment cost and the discounted operation cost over 25 years. Discounting was done with a real interest rate of 3% (EU-Commission, 2012). Due to the reversing price trend in the Nordic electricity market, the average price increase during the past decade has been low (Nordpool, 2016). Thus, an escalation rate of 1% was used in this study.

Table 1
Prices for energy system components

System component	Highest price	Lowest price	Unit
ST (Mauthner, 2016)	605	540	€/m^2
PV (Ahola, 2015)	2000	1400	€/kW
HP (Haahtela and Kiiras, 2013)	325	325	€/kW
Tank (Mauthner, 2016)	1000	815	€/m^3
Borehole drilling (Haahtela and Kiiras, 2013)	33.5	33.5	€/m
Excavation (Rakennuslehti, 2016)	3.0	3.0	€/m^3
Borehole insulation (Rautia, 2016)	88	88	€/m^3
Building quality (Hamdy et al., 2013; Haahtela and Kiiras, 2013)	3000	0	€/house
Pumps (Kolmeks, 2016)	10 000	500	€/kW
Pipes (Taloon.com, 2015)	20	5	€/m

2.3. Optimization method

Optimization generally requires many function evaluations or simulations. When the time required for a function evaluation is long compared to other processes in optimization, it makes sense to reduce the number of required simulations. To do this, artificial neural networks (ANN) were utilized. ANN forms a simple metamodel of a more complicated system. Doing function evaluations with ANN is very fast (microseconds) compared to complete simulation (over ten minutes) and makes optimization more feasible by significantly reducing the time required to complete the optimization. However, training an accurate neural network requires a large amount of samples. Generating these samples requires full simulations and can take as long as or even longer than just conventionally optimizing the system. To bypass this problem, two steps were taken:

1. Instead of a single general neural network, several local neural networks were used. The idea was that neural network accuracy is improved by having to model a smaller range of variables and dealing less with outliers.
2. Instead of generating a complete training set before optimization, the optimization was started with a very small sample set and an inaccurate neural network model, which was updated as optimization progressed. After every iteration, the sample set grew and an updated neural network was generated.

A flowchart of the method progression is shown in Fig. 3. At every iteration, a reference point and a search line were chosen from the objective space. Then, 50% of the current sample set in the objective space was chosen in order of the smallest distance from the search line, using normalized euclidian distance as the metric. These points were used for training the neural network, with 60% selected for the training set, 20% for the validation set and 20% for the test set. At every iteration, optimization was done using the latest ANN. These optimized results from the metamodel were then used as input for the full TRNSYS simulation, to generate exact solutions that were added to the sample set.

Only focusing on points in the same cluster was expected to reduce the effect of outliers and increase ANN accuracy. While similar objective values could be obtained by very different variable values, it was assumed that small changes in decision variables will result in small changes in the objectives. This makes optimization feasible, even though the reverse is not universally true. The method also took advantage of the Achievement Scalarizing Function (ASF) (Wierzbicki, 1980), which can be used to turn the multiobjective optimization problem into a local single-objective problem. The ASF is defined as

$$ASF(x, z, w) = \max_i^M \left(\frac{f_i(x) - z_i}{w_i} \right), \quad (5)$$

where x is the vector of decision variables, z is the reference point located in the direction of optimization and w is the direction of

Table 2
Decision variables for optimization.

Decision variable	Lower bound	Upper bound	Unit	Variable type	Description
A_{ST}	5	40	m^2/building	Continuous	Solar thermal collector area on each roof
V_{hot}	0.5	5.0	m^3/building	Continuous	Hot buffer tank size
V_{warm}	0.5	5.0	m^3/building	Continuous	Warm buffer tank size
A_{PV}	0	60	m^2/building	Continuous	Solar electric panel area on each roof
R_h	0.5	1.5	–	Continuous	BTES height vs. width ratio
V_{BTES}	100	1000	m^3/building	Continuous	Volume of seasonal borehole storage
$d_{insulation}$	0	2	m	Continuous	Thickness of BTES top insulation
α	30	70	°	Continuous	Tilt angle of solar collectors and PV panels
C_{flow}	0	2	–	Discrete	ST flow rate control algorithm
C_{series}	0	2	-	Discrete	ST series mode control algorithm
$E_{building}$	25	50	kW h/m^2/a	Discrete	Space heating demand in buildings
P_{HP}	180	360	kW	Discrete	Heat pump capacity
$N_{boreholes}$	0.05	0.2	boreholes/m^2	Continuous	Borehole density
N_{series}	1	9	–	Discrete	Number of boreholes in series

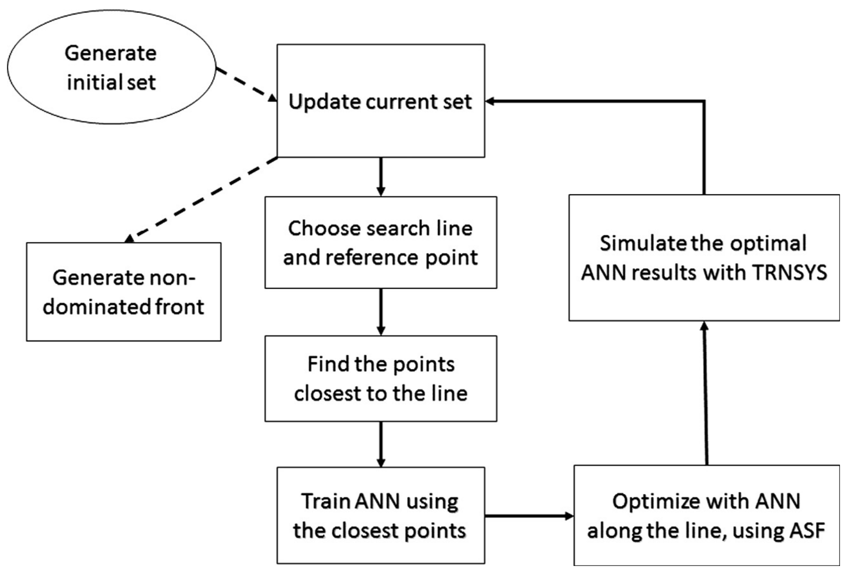


Fig. 3. Flowchart of the optimization process.

the search line (Fig. 4). Additionally, M is the total number of objectives, i is the objective index and f is the vector of objective function values, generated by the ANN. ASF compares the distance of the chosen point from the reference point along each axis, weighed by the search direction w. Now, minimizing the ASF will give a solution that is close to the line and the reference point. Ideally this will be at the intersection of the Pareto optimal (PO) front and the search line, but this is dependent on the accuracy of the neural network model and in practice will not happen until later optimization stages. However, eventually solutions should be generated close to the PO front, increasing the ANN accuracy, allowing the optimization results to improve further. It should be noted that even if the ANN prediction is very inaccurate, the actual simulated solution may still be a good one, because the predictions tend to be overly optimistic while still potentially choosing the decision variables in a sensible manner.

Single-objective optimization was performed with MATLAB's *fmincon* function, which utilizes an interior-point method. The neural network was generated using MATLAB's Neural Network Toolbox. In the ANN, there were one input layer with 14 nodes, one hidden layer with 10 nodes and the output layer with 2 output nodes, the outputs corresponding to each objective function. Each local ANN could be trained and used independently, so parallel

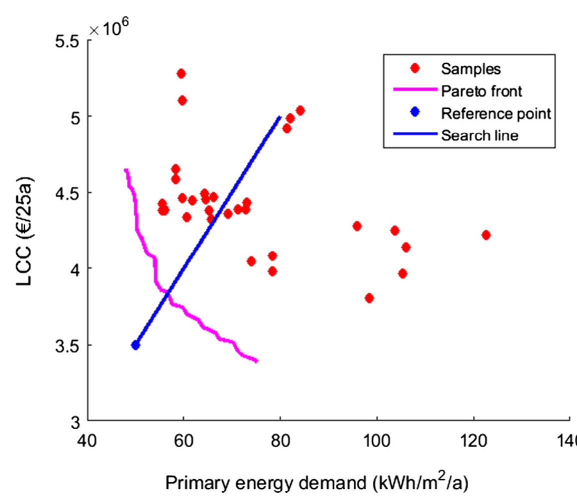


Fig. 4. The search line extends from the reference point towards the known samples, intersecting the Pareto-optimal front.

computation could be utilized to further reduce the time needed for optimization. However, to benefit from the gradual ANN learning, there should not be too many parallel processes. Since each simulation takes around 15 min to run, parallel computation was used in this study.

The ANN prediction error was calculated using the root mean square error

$$RMSE = \sqrt{\frac{1}{N}\sum_{1}^{N}\left(\frac{(f_{\mathrm{ANN},i} - f_{\mathrm{exact},i})}{\Delta f}\right)^2}, \tag{6}$$

To validate the effectiveness of the new algorithm, the optimization was also done using a standard genetic algorithm, NSGA-II (Deb et al., 2002). The crossover probability used for NSGA-II was 0.9, the mutation probability 0.0333, population size 16 and number of generations 80. The GA optimization was performed with the freeware software MOBO (Palonen et al., 2013).

2.3.1. Line selection

One issue with the new optimization method is the selection of search lines. Given a set of known objective values, along which line should the next solution be searched for? The first idea was to take the current non-dominated front and evenly distribute search lines between the extreme points. However, if the extremes are too close, the algorithm may get stuck on a small section and not find the complete non-dominated front. To reduce the chance of this happening, five ranks of non-dominated solutions were found and the most extreme points from all sets were chosen as the outer search lines, with the other lines evenly distributed between them (Fig. 5a). We call this the even distribution method (EDM).

A good Pareto front contains a distribution of solutions from all parts of the front. Therefore, an alternative way for line selection was to choose the lines based on gaps between the non-dominated points. At every iteration, the normalized distances between adjacent non-dominated points were calculated and the optimization lines were placed between the points with the largest distances (Fig. 5b). Ideally, this will form a well distributed PO front and prevent the same point from getting simulated many times, which might happen for lines that never change. To prevent the algorithm from getting stuck in a small region, lower ranks of non-dominated points were used to extend the search space. If there were not enough non-dominated points to choose the desired amount of lines, missing search lines were chosen randomly between the extreme points of the current non-dominated front, extended by 20% outwards. We call this the dynamic distribution method (DDM).

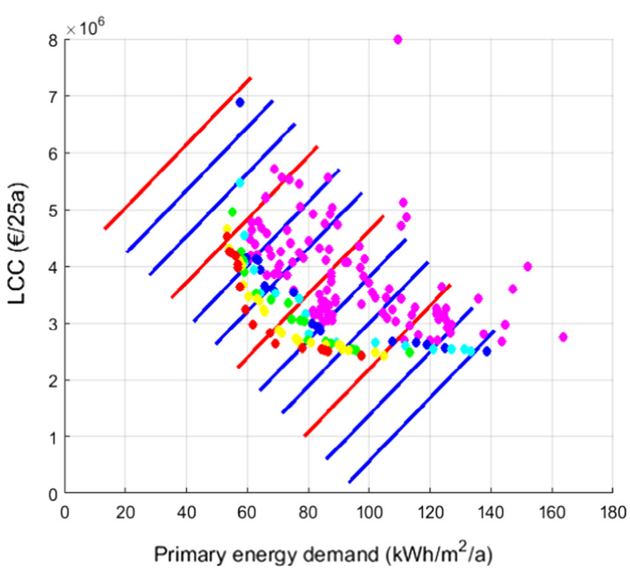
3. Results

3.1. Performance of optimization method

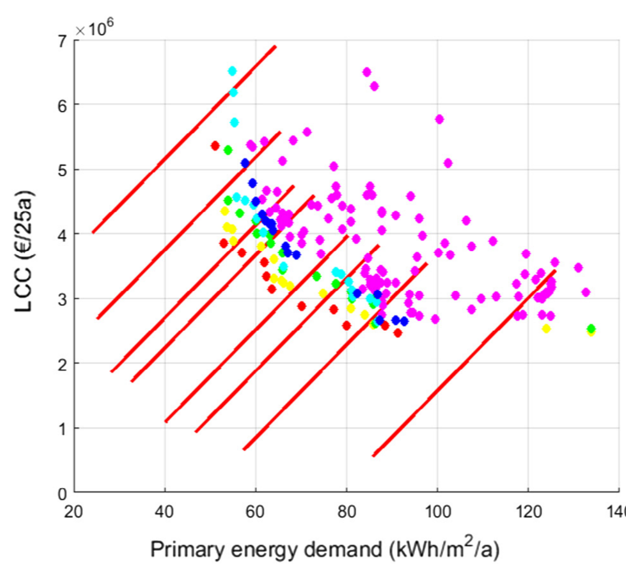
The non-dominated fronts from optimization with both ANN methods and NSGA-II are shown in Figs. 6 and 7. In the early stages, both ANN methods extended the non-dominated front in the bottom-right extreme, compared to NSGA-II. In the end, both ANN methods found better solutions in the middle part of the front compared to NSGA-II, though the dynamic method performed better than the even distribution method overall.

The hypervolume attained with all methods is shown in Fig. 8. The hypervolume was larger for DDM than for NSGA-II, because it extended to the extreme of one objective and found better results in the middle of the PO front. With EDM, the hypervolume sometimes went down, due to critical points getting dominated. Despite the larger hypervolume of DDM, it cannot be said that the ANN method is strictly better than NSGA-II. When the results from both genetic and ANN methods were merged, a better non-dominated front was achieved, as each method found some points dominating those of the other method (Fig. 9).

Fig. 10 shows the root mean square error of the ANN estimations of the objectives vs. the exact values gained from TRNSYS simulations (normalized to the Pareto front width). In the beginning, the error was very large, almost 150%. After adding 200 new samples, the average error had dropped to below 50% and by the end, the error was below 10 and 5%. Prediction improvement was clear, but because the modeled regions changed and the neural networks were retrained after every sample addition, sometimes the prediction accuracy suffered, as shown by the out-



(a) Even distribution method. Line colors indicate which lines are examined in the same parallel iteration.



(b) Dynamic distribution method. Most lines were chosen based on distances in the first non-dominated rank, but two lines were chosen using lower rank solutions to extend the search space.

Fig. 5. Distribution of search lines along the current sample set. Dot colors indicate different ranks of non-dominated solutions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

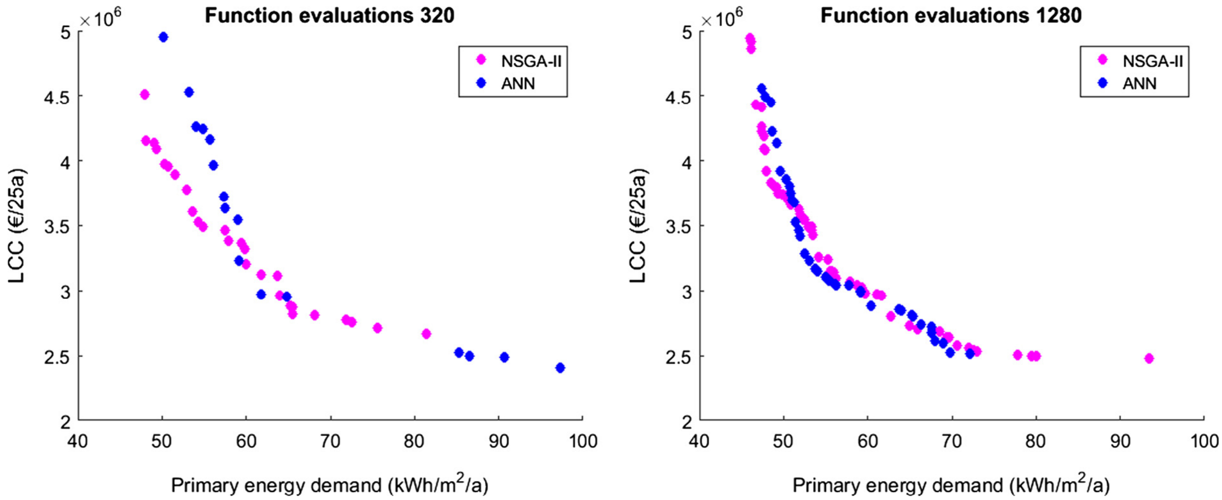


Fig. 6. Non-dominated fronts achieved with the genetic algorithm and the neural network method with evenly distributed lines.

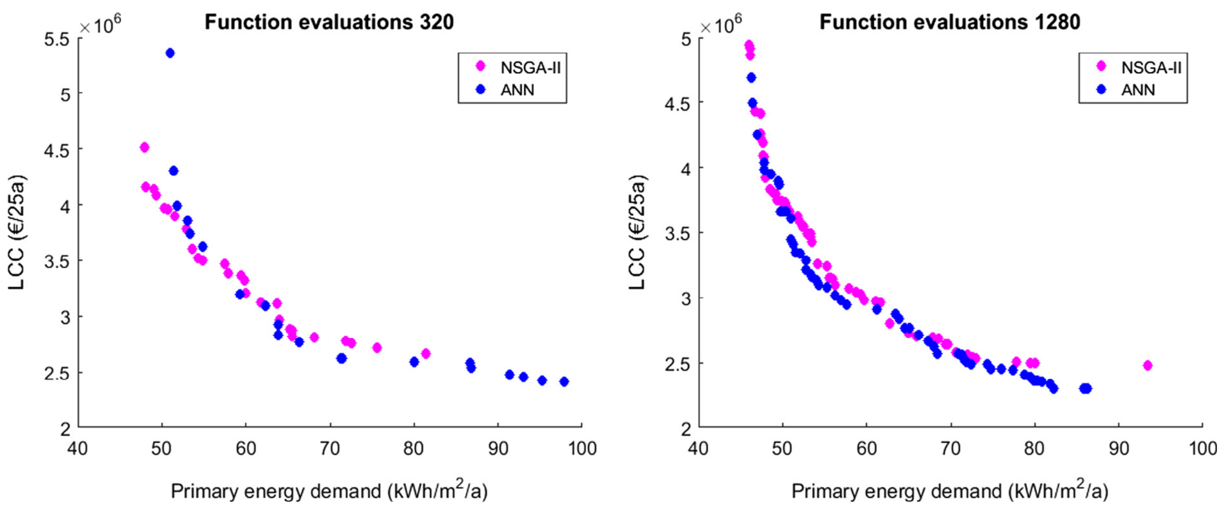
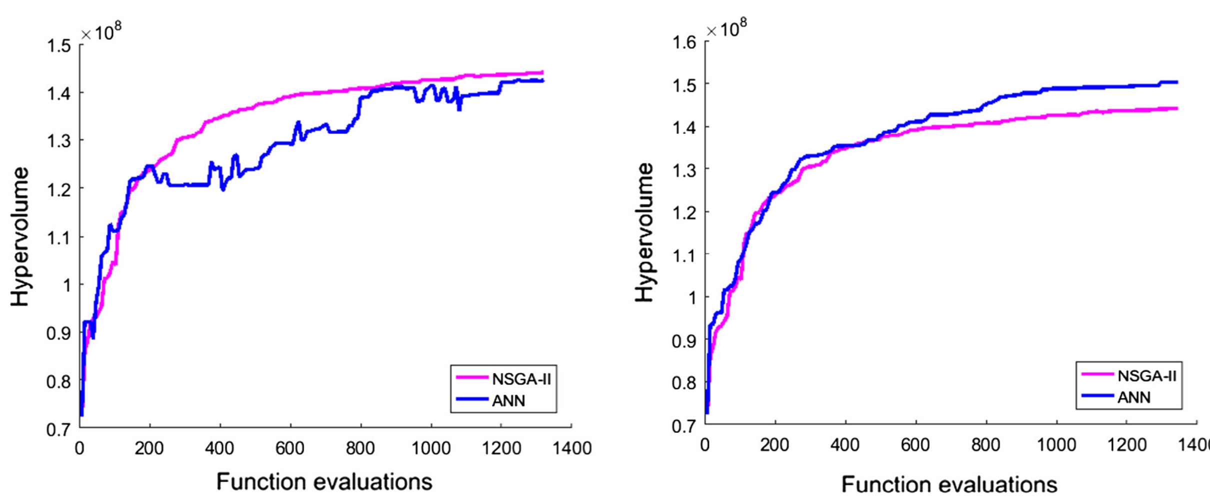


Fig. 7. Non-dominated fronts achieved with the genetic algorithm and the neural network method with dynamically distributed lines.



(a) Even distribution method.

(b) Dynamic distribution method.

Fig. 8. Hypervolume comparison of NSGA-II and both ANN methods.

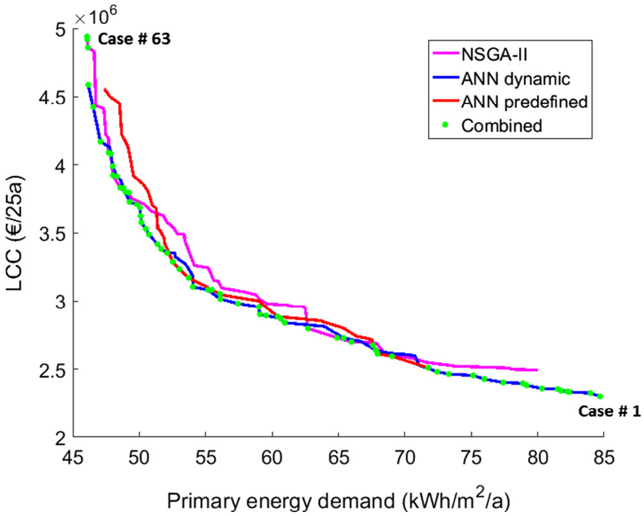


Fig. 9. Non-dominated front generated by the combined results of both the ANN and NSGA-II methods. Each green point corresponds to one case in Figs. 13–16, in order from right to left.

liers in the figure. Despite the general ANN inaccuracy, predictions were correct enough to allow optimization to proceed.

3.2. Features of the optimized solar community

All the results here refer to the combined non-dominated front shown in Fig. 9. A detailed look at the system performance during two typical summer and winter days is given in Fig. 11 (optimal case #59). It shows how the warm tank stored solar energy and transferred it to the BTES during summer and how little solar energy was available in winter. The BTES was discharged to the warm tank through the bypass to provide space heating, while the heat pump was used for additional heating of both the warm and hot tanks when their temperatures dropped below the setpoints.

For the optimal solutions, the initial investment cost was 56–92% of the total LCC, with the rest going to operation costs. The distribution of investment costs is shown in Fig. 12. The solar thermal collectors formed the biggest part of the investment, followed by

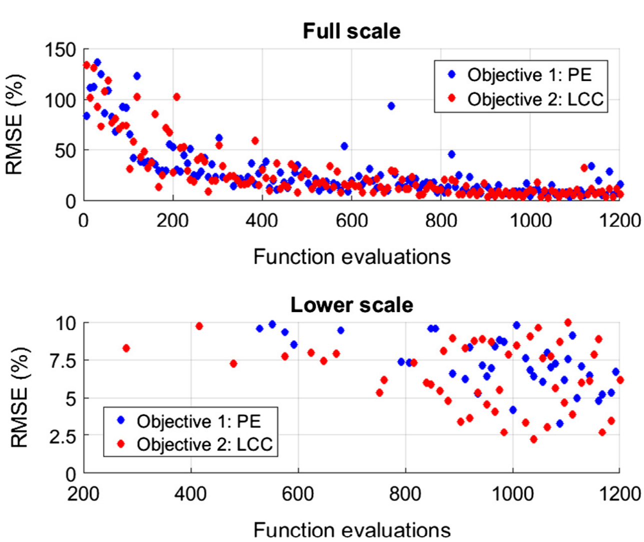


Fig. 10. Normalized root mean square error (RMSE) of ANN metamodel prediction vs. TRNSYS simulation for each set of 8 new solutions. Normalization was done according to the final Pareto front widths in objective space.

the PV system. BTES size was tied to the solar collector area. When ST collector area was below 10 m^2, the BTES volume was very large (130 000–200 000 m^3), because without enough solar recharging, the HP heat drain would cool the ground to temperatures unusable by the HP. In these cases there were only 0.5–0.75 boreholes per building (about 0.05 boreholes per m^2), but the boreholes were deep, with height-to-width ratio of 3. For collector areas of 15–25 m^2, the BTES had minimal volume (10 000 m^3), maximal width (0.5 height-to-width) and high borehole density. The number of boreholes per building varied from 0.75 to 1.4 and the borehole density from 0.1 to 0.2. Finally, with collector areas above 25 m^2, the BTES volume increased to the range of 30 000–50 000 m^3 with the height-to-width ratio varying between 0.5 and 1.1. The borehole densities were between 0.15 and 0.2, with 1 to 3 boreholes per building. BTES top insulation was only needed when each house had more than 25 m^2 of collectors.

Thus, if there is little energy to store, the optimal BTES shape is deep and narrow like in conventional heat pump boreholes, allowing environmental heat to recharge the system. With more solar heat input, the BTES needs to be wider, to allow retention of heat stored in the center. In the optimal solutions, boreholes were strongly connected in series. Most common configuration had 9 boreholes connected in series, which was the maximum allowed setting in the model. The stronger the seriality, the larger part of solar heat is injected in the center of the storage. When adding more collectors, the borehole density should also be increased.

With high solar thermal and storage capacity, the need for the heat pumps was reduced significantly. With the best performing system, the seasonal performance factor (SPF) reached 21. The solar fractions of heating and total electricity use are shown together with the corresponding amount of ST and PV panels in Fig. 13. The heating solar fraction varied between 64 and 95%, meaning that the goal of over 90% solar-based heating was achieved. PV panels took up a significant amount of the roof area in all cases, due to the possibility to sell excess power back to the grid. Once the roof was full, improvement to heating solar fraction happened by replacing PV panels with ST collectors. In all cases, the discounted value of lifetime PV generation was roughly equal to the PV investment cost. When appliance electricity demand was included, the total solar fraction was between 24 and 39%. Because no electricity storage was utilized, installing more than 3 kW of PV panels did not increase self-consumption, saturating the maximum total solar fraction to 39%. The total imported electricity was 50 kW h/m^2 in the worst case and 27 kW h/m^2 in the best case.

The distribution of electricity use in different cases is shown in Fig. 14. The need for external energy for heating was reduced from 25 kW h/m^2 to 3 kW h/m^2. The reduction was caused by improved building efficiency and larger solar thermal energy generation (Fig. 15). Solar energy was used to meet heating demand directly, but also to increase the ground temperature to higher values. When the ground temperature was high enough, the heat pump bypass could be activated, which allowed direct use of the seasonally stored heat, without utilizing the heat pump. When the solar collector area was less than 20 m^2 per house, the seasonal storage never heated enough to bypass the HP when heating. Regardless, improved HP efficiency from heated ground minimized the need for electricity. As the amount of solar collectors increased, so did the losses. BTES system efficiency went down as higher ground temperatures (through larger collector areas) were attained. Above 30 m^2 per house, the BTES efficiency was 40%, at which point most of the stored energy was lost. All optimal combinations used the maximally efficient building, which indicates that it is more economical to reduce heat consumption than increase generation.

The tilt angle to maximize total annual solar energy generation in Finland is 40°, but the average tilt angle obtained by the opti-

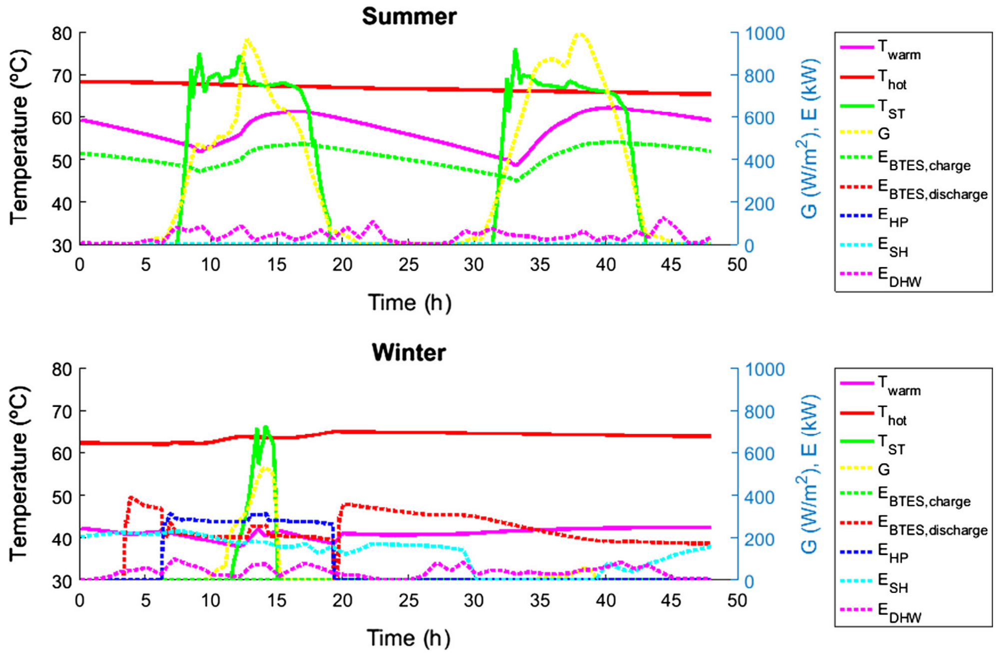


Fig. 11. Performance of the energy system during two days in summer and winter. Shown are the temperatures at the top of the warm and hot tanks and the output temperature of the solar thermal system. Also visible are the solar insolation G, the thermal energy transfer to and from the BTES, the heat generated by the heat pump and the SH and DHW energy demand. These profiles are from optimal case #59 of Fig. 9.

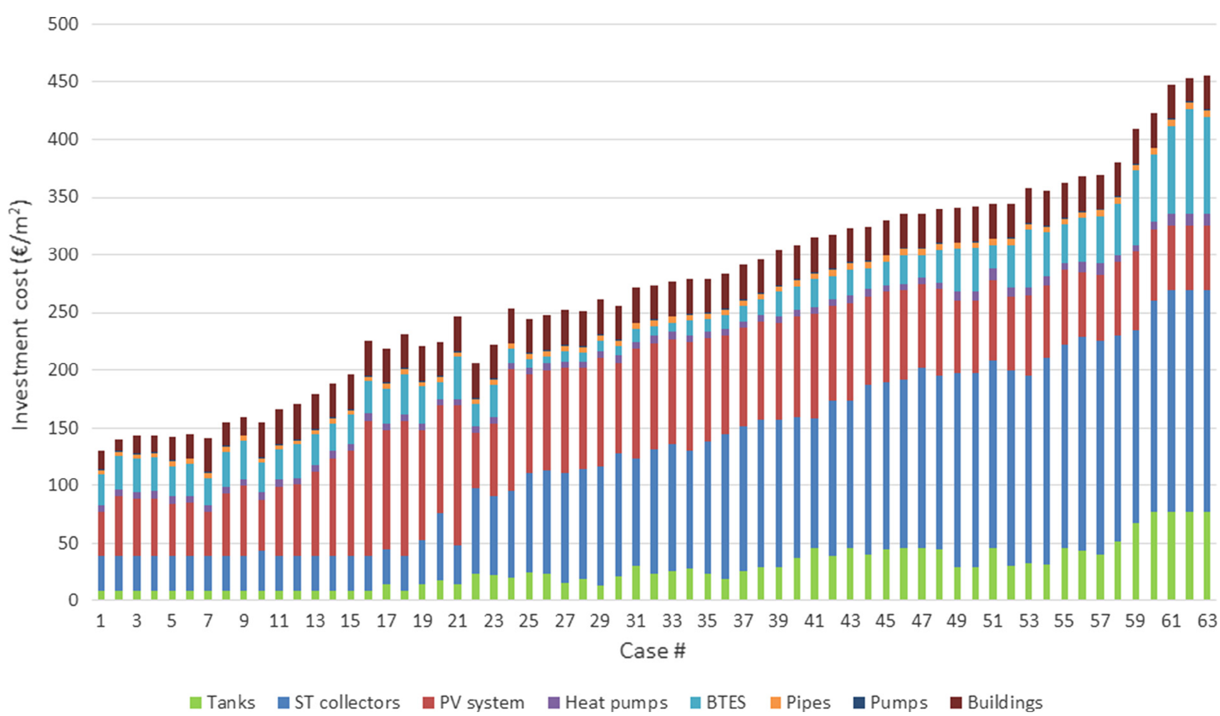


Fig. 12. Distribution of initial investment cost between different components. The cost is given per house floor area. Case numbers refer to the points in the combined non-dominated front in Fig. 9.

mization was 50°. This higher angle increases direct use of solar energy during spring and fall, while reducing overproduction in summer, when the demand is lower. In all the optimal solutions, the collectors were connected to the tanks in parallel, instead of in series. This way, DHW level temperatures were not used for SH or preheating purposes, improving efficiency.

It is noteworthy that the final optimal solutions obtained by the dynamic ANN method were closer to the variable boundaries than with NSGA-II. Figs. 16 and 17 show the properties of the BTES system obtained by both methods. Most obvious difference is in the BTES volume. The dynamic method found solutions where a small ST area was combined with a very large BTES volume (over 150000 m^3) and a small amount of deep boreholes, which allowed the use of heat pumps for heating without draining the ground of all energy. NSGA solutions with similar objective values had BTES volumes less than 50000 m^3, but employed more solar thermal

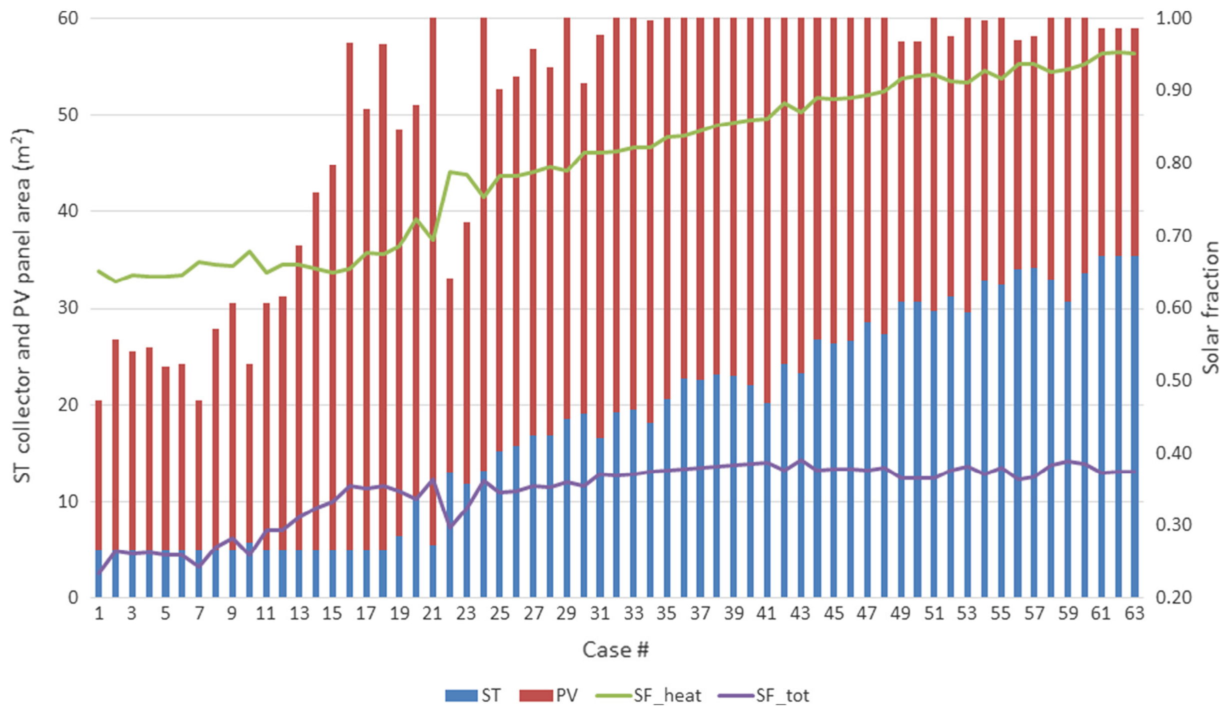


Fig. 13. The amount of rooftop area (per building) used by ST and PV panels for each case, along with the solar fractions for heating and total electricity. Case numbers refer to the points in the combined non-dominated front in Fig. 9.

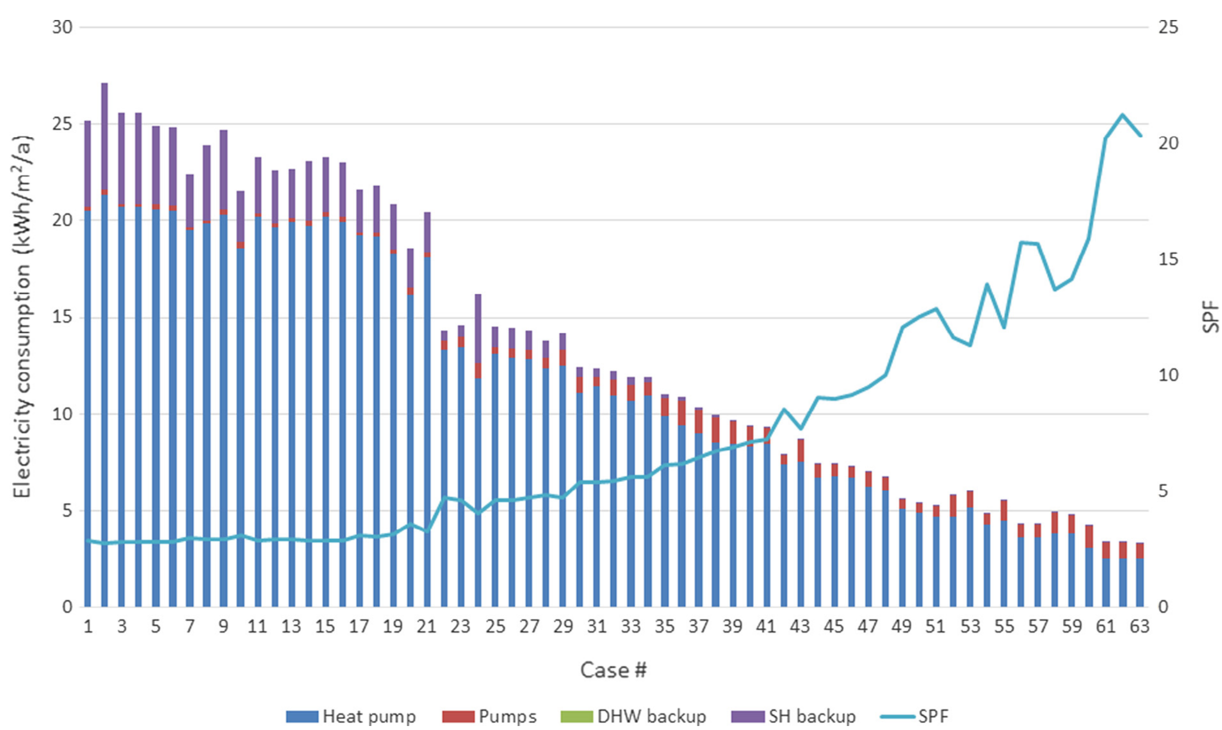


Fig. 14. Electricity consumption of heating by different components in the energy system. Case numbers refer to the points in the combined non-dominated front in Fig. 9.

collectors to recharge the ground. For moderate ST areas, the ANN method found solutions with minimal BTES volume (10000 m^3), large borehole densities, minimal hot tank volume and larger warm tank volume. Solutions in the same region found by NSGA had larger BTES volumes while having the hot and warm tanks be almost equal size to each other. The ANN algorithm was able to find better solutions at the decision boundaries, which the genetic algorithm was unlikely to reach.

4. Discussion

4.1. Optimization method

The algorithm might be improved by saving an archive of good neural networks. If the currently examined line is close to an archived line with accurate predictions, reusing that ANN might give good predictions for this region as well.

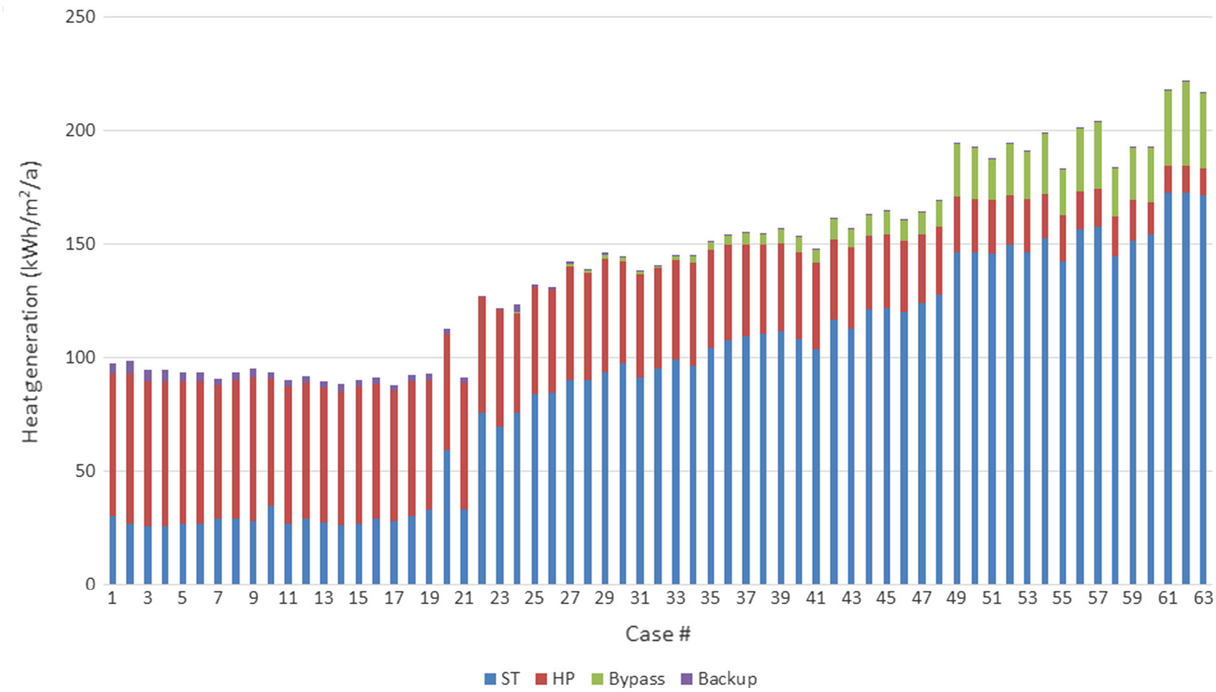
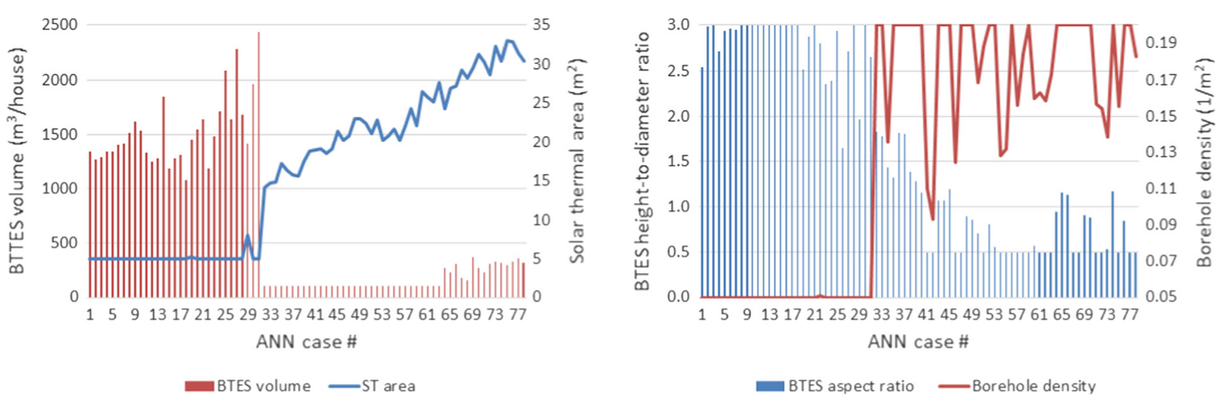


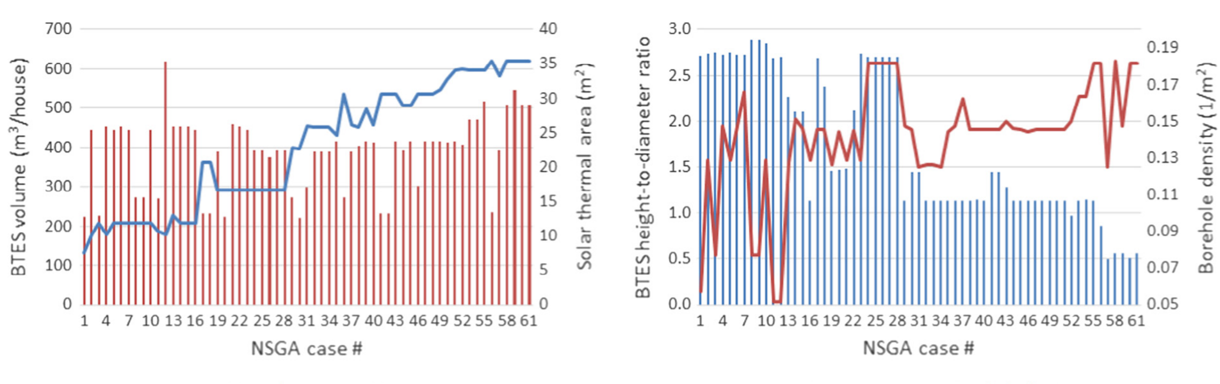
Fig. 15. Heat generation by different components in the system. Case numbers refer to the points in the combined non-dominated front in Fig. 9.



(a) BTES volume and ST area.

(b) BTES aspect ratio and borehole density.

Fig. 16. BTES properties in the non-dominated front obtained by the dynamic ANN method.



(a) BTES volume and ST area.

(b) BTES aspect ratio and borehole density.

Fig. 17. BTES properties in the non-dominated front obtained by NSGA-II.

Additional work should be done to limit the time spent on finding solutions near the edges of the Pareto front. Extending the non-dominated front may cause large gaps, which cause the dynamic method to focus on the extreme region, which can reduce optimization efficiency in the central part of the Pareto front. Often the outer edge solutions are of little practical interest, due to the small gains in one objective at large cost to another. However, it is important to be able to search outside the current non-dominated front, to prevent the algorithm from getting stuck in a small region of the actual front.

Adjusting the number of gaps examined at each iteration has an effect on DDM performance. With a small number, the algorithm may get stuck if new solutions are not found in the largest gaps. With a large number, getting completely stuck becomes unlikely, but more focus will be given to smaller gaps, possibly wasting time on a fully optimized region.

It should be noted that even when the ANN prediction accuracy is extremely bad the iteration may still result in an improved solution, because the selected decision variables can be close to an optimum despite the inaccuracy in the objective space. The fraction of samples used for ANN training may have an influence on the results. Currently 50% were used, but a larger fraction might be beneficial. Starting at a high fraction and reducing it as the number of samples increases might also be useful.

Changing the neural network architecture might improve the accuracy of the metamodel and thus make the algorithm converge faster to the Pareto optimal front. In this case, the ANN outputs were the final objective functions. Modeling the different subcomponents of the final objectives could make it easier to train the ANN and thus improve metamodel accuracy.

Different line selection methods may work better for different problems. While the dynamic line selection worked better in this case study, it may fare poorly in discontinuous problems where the PO front has infeasible sections which leave large gaps between PO solutions. The even distribution method might be better suited for these problems.

Compared to the genetic algorithm, the ANN method was more likely to find solutions at the boundaries of the variables. For example, many solutions obtained by the ANN method had minimum values for the hot tank and BTES volumes. This highlights the fact that solutions with similar objective values may obtained by very different parameters. This can be important for a decision maker, as practical conditions might favor one solution over another equally good solution, even if there is no hard constraint. Thus, to ensure larger diversity, it might be useful for an optimization algorithm to rank solutions not just by their objective function values, but also by their variable values, through some kind of similarity factor.

4.2. System design

Most optimal solutions had large enough solar electric capacities that more solar electricity was sold off-site than actually used in the community. Above 3 kW capacity per building, the benefit for self-consumption was minimal, but due to solar energy exports an economical balance could be achieved. More roof space was dedicated to PV panels in the low performance cases (higher energy demand), while ST collectors dominated in the high performance cases.

The BTES system was significantly insulated only in the very high solar fraction cases. With small ST capacity, the temperature of the ground remained low and thus no insulation was needed to prevent heat losses. Borehole density was tied to system performance and solar thermal area. The lower the ST area, the lower the borehole density. When the ST area was low, the BTES volume was very high, while the borehole density was minimal. Without a balancing solar energy injection, a small ground volume got drained of thermal energy, rendering the heat pumps inoperable.

For larger ST systems, the BTES volume was first reduced, but then started increasing again with more ST collectors. For a given aspect ratio, increasing BTES volume should always improve efficiency, due to storing a larger part of the energy in the center, far away from the outer edge loss surface. This makes BTES systems excellent for scaling up. However, the storage volume has to be in balance with the generation capacity to ensure meaningful temperature rise. Heat pumps for single buildings usually operate through a single deep borehole. The deep and narrow "storage" allows natural regeneration of thermal energy. Wide borehole fields reduce heat loss from the center to the environment, but they also slow down natural regeneration. Thus, large centralized GSHP systems require some means of forced regeneration. If solar heat is not available, running the heat pumps in cooling mode would help, if cooling needs were significant. A BTES system could be connected to a conventional district heating grid for regeneration. Excess heat from cogeneration plants during low consumption hours could be stored, as well as waste heat from industrial processes.

With a moderate amount of ST collectors the seasonal heating performance was already high, which reduced the gains from additional increase in ST area. Thus, significant increase in electricity price would be needed to make larger ST systems feasible. If more roof space was available, such as on external garages, the space would likely be taken up by more PV panels. However, having a large continuous roof could lower solar collector installation costs. As solar collectors are the most expensive component in the system, any price reductions could have a significant effect on the LCC. If selling of excess electricity was not possible, the amount of PV panels would be reduced, never going above 4 kW per building.

5. Conclusions

A new optimization method based on simultaneous ANN training and system optimization was developed and tested on a solar community design problem. The new method was able to solve the minimization problem and partially outperform NSGA-II. The ANN method was able to find better solutions in some sections of the non-dominated front and attained a larger hypervolume. Thus, the new method shows potential in solving problems where function evaluation takes a long time compared to the actual optimization process. The ANN method easily allows for parallel computation.

This study has served as a proof of concept for the new optimization method, but further work is needed to improve the line selection algorithms, for example, with respect to expanding the search outside the current non-dominated front. Constraint handling also needs more research, as discontinuous problems may pose difficulties for the algorithm. A detailed analysis of the method using standard test problems will be made in a future study.

The most important finding from the solar community case study was that even in a high latitude Nordic country it is possible to cover 95% of heating demand through solar energy. However, achieving such high solar fractions is very costly, requiring investments as high as 49 000 €/house. Still, an SF of 80% was achievable with the much lower cost of 30 000 €/house. Solar thermal collectors were the largest single cost component in the system. In a competitive business environment these prices are likely to go down. Until then, to make these systems competitive would require some government intervention in the form of investment incentives or increased carbon prices. As EU climate change targets require drastic reduction of energy demand, high solar fraction heating systems may become a very important technology despite the current cost.

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